MILITARY SPECIFICATION

FUZE, PIBD, XM740, SECOND ENVIRONMENT SENSOR
ELECTRONIC MODULE FOR

This amendment forms a part of Military Specification MIL-F-63231 (AR) dated 17 August 1979, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title block: Add "(M740)" to title and wherever it appears in this specification.

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3.2.1.1: Delete in its entirety and substitute the following:

"3.2.1.1 Output signals. The SESE shall provide a "go" (arm) signal or a "no go" (not arm) signal, depending upon the condition and timing of the input signals. The number 1 input signal shall be a single pulse with a voltage rise from 0 to 16 volts in 80+ 5, -0 microseconds and a decay to less than 10 volts at 100+ 10 microseconds and shall be applied to terminals 1 and 2 with terminal 1 positive and terminal 2 negative. The number 2 input signal shall be a single pulse with a voltage rise from 0 to 16 volts in 40 + 5 microseconds and shall be applied to terminals 3 and 4 with terminal 3 positive and 4 negative."

3.2.1.1.1: Delete in its entirety and substitute the following:

"3.2.1.1.1 Go Output signals. The SESE shall produce a "go" signal when input signal number 2 is applied within 180 microseconds or less after the application of input signal number 1, as measured from the beginning of each input signal pulse. A "go" signal shall be a pulse which changes from less than +0.2VDC and maintained at greater than 4.0VDC for 0.7 seconds minimum. This signal shall be measured at terminals 6 and 5 with terminal 6 positive and terminal 5 negative. The "go" signal shall occur within 40 microseconds after the start of input signal pulse number 2."

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3.2.1.1.2: Delete in its entirety and substitute the following:

"3.2.1.1.2 No-Go Output signals. The SESE shall produce a "no go" signal when input signal number 2 is applied at 270 microseconds or more after the application of input signal number 1, as measured from the beginning of each input signal pulse. A "no go" signal shall be a voltage of not greater than 0.2VDC. This signal shall be measured at terminals 6 and 5 with terminal 6 positive and terminal 5 negative. The "no go" signal shall be maintained continuously from the application of signal number 1 until the SESE is discharged."

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Figure 1: Delete and substitute the following:

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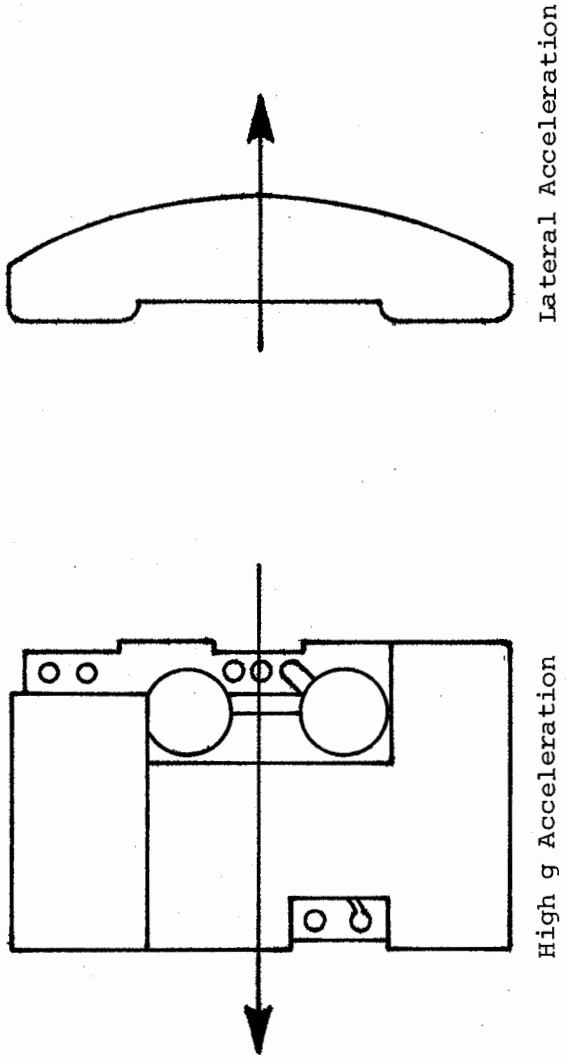


Figure 1 - Direction of Applied Acceleration

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4.5.2.1: Delete in its entirety and substitute the following:

"4.5.2.1 Go Output signal. Apply input signals number 1 and number 2 to proper terminals of the SESE as defined by 3.2.1.1 and with the polarity as specified by 3.2.1.1, and with input signal number delayed 180 + 10, -0 microseconds after input signal number 1, as measured from the beginning of each input signal pulse; the output measured at terminals 5 and 6 with an instrument of 10 ± 2% meg ohm input impedance across a 1.1 ± 1% meg ohm resistor shall be as specified in 3.2.1.1.1."

4.5.2.2: Delete in its entirety and substitute the following:

"4.5.2.2 No Go Output signal. Apply input signals number 1 and number 2 to proper terminals of the SESE as defined by 3.2.1.1 and with the polarity as specified by 3.2.1.1, and with input signal number 2 delayed by 270 + 0, -10 microseconds after input signal number 1, as measured from the beginning of each input signal pulse; the output measured at terminals 5 and 6 with an instrument of 10 ± 2% meg ohm input impedance across a 1.1 ± 1% meg ohm resistor shall be as specified in 3.2.1.1.2."

4.5.7: Delete in its entirety and substitute the following:

"4.5.7 Environmental tests. With the exception of high and low temperature tests the SESE shall be mounted in a carrier or a test fixture which simulates the mounting of the SESE by its normal means in its normal position in the projectile."

4.5.7.1: Delete in its entirety and substitute the following:

"4.5.7.1 High temperature. The SESE shall be conditioned at 145 +5, -0°F for a minimum of one hour. While at the high temperature, the units shall be tested as specified in 4.5.2. The tests may be performed outside the temperature chamber if accomplished within one minute after removal from the chamber."

4.5.7.2: Delete in its entirety and substitute the following:

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"4.5.7.2 Low temperature. The SESE shall be conditioned at -25 +0, -5°F for a minimum of one hour. While at the low temperature condition, the units shall be tested as specified in 4.5.2. The tests may be performed outside the temperature chamber if accomplished within one minute after removal from the chamber. NOTE: Care must be taken to prevent frost from forming on the device which will affect the testing."

4.5.7.3, lines 1 and 2: Delete the first sentence in its entirety and substitute the following:

"The SESE shall be mounted in a carrier or test fixture as specified in para. 4.5.7."

line 3: Delete "±5°F" and substitute "+5, -0°F".

lines 6, 7, and 8: Delete "Remove the ... in 4.5.2" in its entirety and substitute the following:

"Remove the carrier/SESE assembly from the test projectile. Remove the SESE from the carrier and condition it at +145 +5, -0°F for one hour. While at the high temperature, the units shall be tested as specified in 4.5.2. The tests may be performed outside the temperature chamber if accomplished within one minute after removal from the chamber."

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4.5.7.4, lines 1 and 2: Delete the first sentence in its entirety and substitute the following:

"The SESE shall be mounted in a carrier or test fixture as specified in para. 4.5.7."

line 3: Delete "±5°F" and substitute "+0, -5°F".

lines 6, 7, and 8: Delete "Remove the ... in 4.5.2" in its entirety and substitute the following:

"Remove the carrier/SESE assembly from the test projectile. Remove the SESE from the carrier and condition it at -25 +0, -5°F, for one hour. While at the low temperature, the units shall be tested as specified in 4.5.2. The tests may be performed outside the temperature chamber if accomplished within one minute after removal from the chamber. NOTE: Care must be taken to prevent frost from forming on the device which will effect the testing."

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Major 101: Delete "Gage" and substitute "Gage/4.5.3".

Majors 102, 103, 104, and 105: Delete in their entirety and substitute the following:

"102 Go signal initiation/output
100% 3.2.1.1.1 4.5.1/4.5.2

103 No go signal initiation/output
100% 3.2.1.1.2 4.5.1/4.5.2"

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Major 101: Delete "Gage" and substitute "Gage/4.5.3".

Majors 102, 103, 104, and 105: Delete in their entirety and substitute the following:

"102 Go signal initiation/output
100% 3.2.1.1.1 4.5.1/4.5.2

"103 No go signal initiation/output
100% 3.2.1.1.2 4.5.1/4.5.2"

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Major 102: Delete in its entirety and substitute the following:

"102 Conformal coating; 100% 3.3.2 4.5.4".

Custodian:
Army-AR

Preparing activity:
Army-AR

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